Riverside Community College - Department of Computer Science
CSC 220 Data Structures in Java 4 credits - Course Syllabus
Prerequisite: CSC 110 Programming Fundamentals with a grade of C or better.
Course description: abstract data types and their implementation in Java. Lists, stacks, queues, trees, binary search trees, heaps, hash tables and graphs. Recursion, generics, iterators and an introduction to algorithm analysis with big-O notation.
Learning outcomes: implement and compare linked and array-based structures; analyse the running time of searching and sorting algorithms; choose an appropriate data structure for a problem; write and test object-oriented Java programs of moderate size.
Weekly topics: 1 Java review and generics; 2 algorithm analysis; 3-4 linked lists, stacks and queues; 5 recursion; 6-7 trees and binary search trees; 8 heaps and priority queues; 9 hash tables; 10-11 graphs, BFS and DFS; 12 sorting algorithms; 13 review; 14 final exam.
Contact hours: 3 hours lecture and 2 hours lab per week for 15 weeks.
Assessment: programming projects 40%, quizzes 15%, midterm 20%, final exam 25%.
Synthetic document created for the BU transfer credit demonstration.